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AI Evaluation of Pairwise Responses Using the Preference-Dissection Dataset

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Tools & Technologies:

Python (Pandas, Numpy) | Matplotlib/Seaborn| Jupyter Notebook | GitHub

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1. Summary

This project evaluates paired AI responses using the Preference-Dissection dataset to determine which of two outputs is stronger, why it is stronger, and which dimensions most influence comparative preference. Unlike single-response evaluation, this workflow requires both responses to be scored independently and then compared through a structured preference decision.

The final chart outputs show that the overwhelming majority of evaluated pairs were judged as ties, with only a small number favouring response_1 and none favouring response_2. This means the two compared responses were usually similar in overall quality rather than clearly separated. When differences did appear, they were driven mainly by clarity and completeness, while the weaker responses were most often associated with missing key steps and weaker reasoning rather than factual failure.

2. Business/Analytical Question

The business and analytical questions the evaluation addressed are:

- Which of two AI responses is better, and why?
- What quality dimensions most influence preference decisions?
- What recurring failure types appear in the weaker response?
- How consistent are scoring decisions across similar prompts?

3. Dataset

Dataset used: Preference-Dissection dataset parquet/CSV extract. The source dataset is suitable for pairwise evaluation because it contains a query together with response_1 and response_2 in a preference-oriented structure.

4. Tools Used

- Python (pandas, numpy) for data loading, scoring, and aggregation
- Jupyter Notebook for side-by-side evaluation workflow
- Matplotlib and seaborn for chart generation and table outputs
- CSV and PNG exports for top closest cases and visual reporting assets

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5. Evaluation Process (What I Did)

1. Loaded the preference dataset and extracted usable text for query, response_1, and response_2.
2. Built a pairwise evaluation data frame with separate score columns for both responses across accuracy, relevance, completeness, consistency, clarity, and reasoning quality.
3. Calculated total scores for response_1 and response_2 and assigned a preferred response label of response_1, response_2, or tie.
4. Recorded weaker-response error labels and added structured justification fields explaining why one response was stronger or why both were effectively tied.
5. Computed score gaps to identify close or disagreement cases and used a Top 10 Closest Cases table to support analysis where strict disagreement thresholds returned almost no examples.

6. Key Visualizations

1. Preferred Response Distribution: The chart shows a strong concentration of tie outcomes, a small number of cases favouring response_1, and no visible cases favouring response_2. This indicates that most evaluated response pairs were judged to be very similar in quality rather than strongly differentiated.
2. Average Quality Dimension Gap Between Responses: The chart shows that clarity has the largest average gap, followed by completeness. Accuracy, relevance, consistency, and reasoning quality have smaller average gaps. This means that most preference differences are driven by how clearly and fully the response is expressed rather than by basic correctness.
3. Recurring Failure Types in Weaker Responses: The weaker-response error taxonomy shows only two major recurring issues in the visible chart: missing_key_step_or_omission and weaker reasoning. This indicates that the losing responses are typically weaker because they are incomplete or shallow rather than wrong.
4. Top 10 Closest Cases Table: The closest-case table shows multiple tie examples and very similar total scores between response_1 and response_2. Several rows also contain noisy or low-quality text, which helps explain why the system often could not strongly differentiate between the two responses.

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7. Key Insights

1. The evaluation produces a very high rate of tie decisions, which suggests that the paired responses are usually similar in quality rather than clearly superior or inferior to one another.
2. Preference decisions are driven mainly by clarity and completeness. These are the dimensions with the largest average gaps, which means they are the strongest practical differentiators in the current subset.
3. The weaker responses do not fail mainly because of factual error or irrelevance. Instead, they lose because they omit important steps or show weaker reasoning quality.
4. The very low number of close/disagreement cases under strict thresholds indicates that the current rubric tends to produce either ties or decisive separations rather than many borderline comparisons.
5. The closest-case table suggests that noisy or degraded text quality may contribute to high tie rates, reducing the system's ability to distinguish clearly between alternative responses.

8. Business Implications and Recommendations (Why This Project Matters)

From a business perspective, pairwise preference evaluation is valuable because many AI systems do not fail in obvious ways. Instead, they produce multiple plausible answers where the practical question is not “is this wrong?” but “which answer is more useful?”. This project shows that in the current subset, many competing responses are difficult to distinguish strongly, which means the model may be producing outputs that are broadly acceptable but insufficiently differentiated.

This matters for ranking systems, assistants, and decision-support workflows because if two responses are usually tied, the system may struggle to consistently prioritize the most helpful answer. The chart evidence suggests that improving clarity and completeness would likely increase differentiation and make preference decisions more meaningful. Likewise, reducing omissions and strengthening reasoning should improve the weaker responses more effectively than focusing narrowly on factual correctness.

- Prioritize improvements in clarity and completeness, as these are the strongest observed drivers of preference differences.
- Target missing steps and weaker reasoning in the weaker responses, since these appear to be the dominant recurring failure patterns.

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- Use pairwise evaluation as a ranking and optimization tool rather than only as a pass/fail quality check.
- If stronger disagreement analysis is required, expand the evaluation to a broader or more ambiguous subset where more borderline comparisons are likely to appear.

9. Challenges and How to Solve Them

- Challenge: Response pairs may be too similar, resulting in many ties and limited disagreement analysis.
Solution: Report tie dominance transparently and supplement with a Top 10 Closest Cases table for richer explanation.
- Challenge: Pairwise scoring requires double scoring effort because both responses must be evaluated independently.
Solution: Use a structured scoring template and semi-automated first-pass scoring where appropriate.
- Challenge: Quality differences can be subtle and spread across several dimensions at once. Solution: Use dimension-gap analysis to identify the most influential factors and require written justifications for preferences.
- Challenge: Noisy or malformed response text can weaken interpretation quality.
Solution: isolate noisy cases explicitly and treat them as part of the explanation for high tie rates or weak differentiation.

10. What I would Improve Next

To make the project even stronger in an evaluation setting, I would:

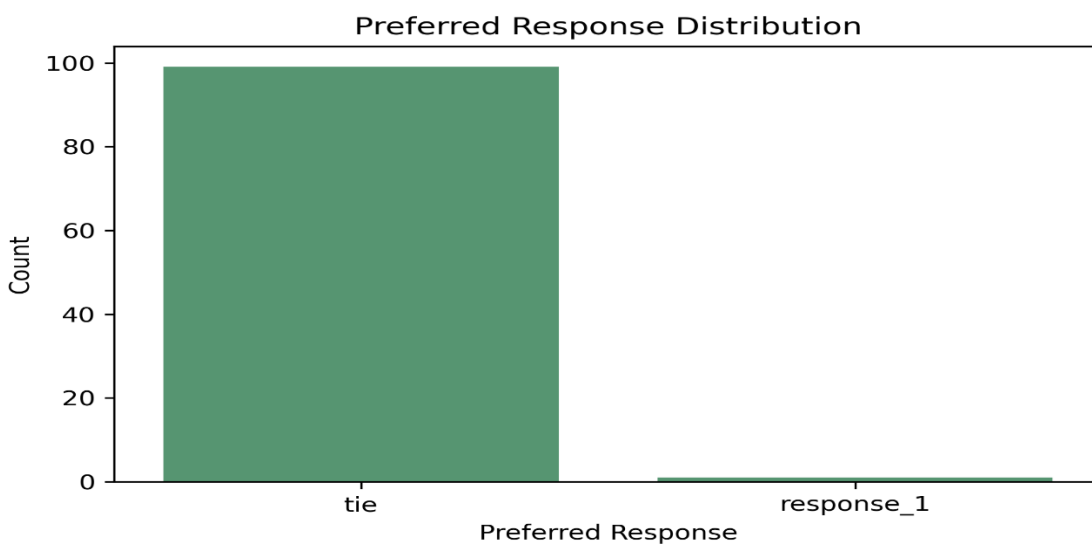
- Expand the scored subset to include more difficult or ambiguous pairs so that disagreement-pattern analysis becomes richer.
- Compare evaluator preference choices against any available human preference labels in the dataset to validate alignment.
- Add inter-rater agreement checks to test whether pairwise preference decisions are consistent across reviewers.
- Create a standard Top Closest Cases view whenever strict close-case thresholds return too few examples.
- Refine weaker-response error labels so they map more directly to completeness, clarity, and reasoning failures.

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11. Files and Links

- Notebook: Dataset 3 pairwise preference evaluation notebook (.ipynb)
- Visual outputs: pref_preferred_response_distribution.png, pref_dimension_gap_distribution.png, pref_weaker_response_error_taxonomy.png, pref_top10_closest_cases_table.png
- Tabular outputs: pref_top10_closest_cases.csv

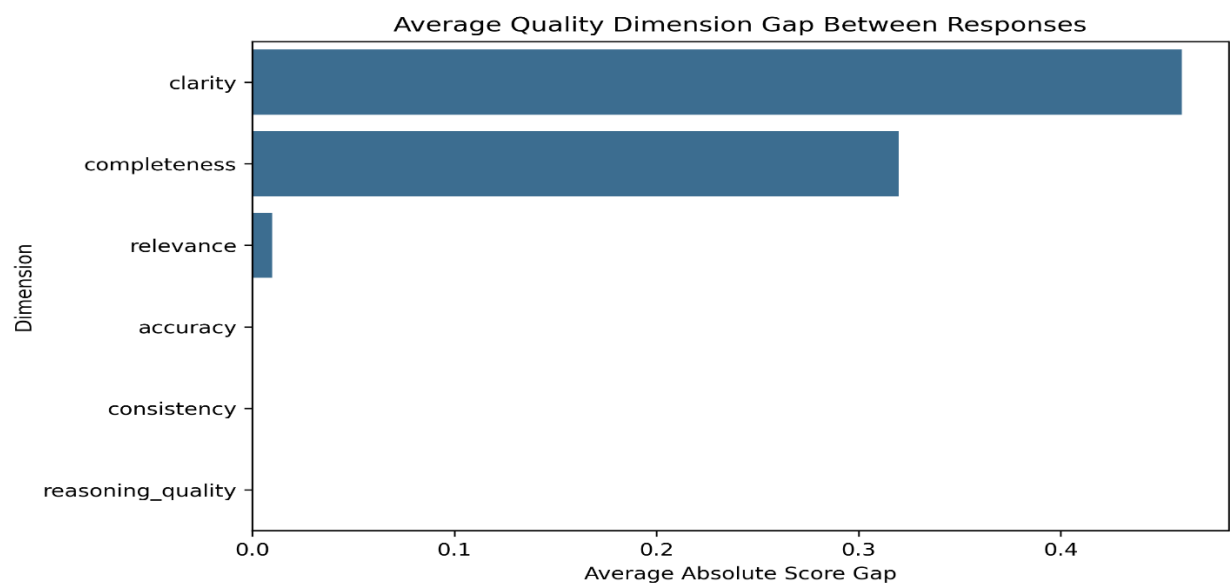
Chart 1: Preferred Response Distribution



Insight: The distribution is heavily dominated by **tie outcomes**, with only a small number of cases favouring Response 1 and none favouring Response 2. This indicates that most paired responses are judged to be of **similar quality**, suggesting limited differentiation between competing outputs.

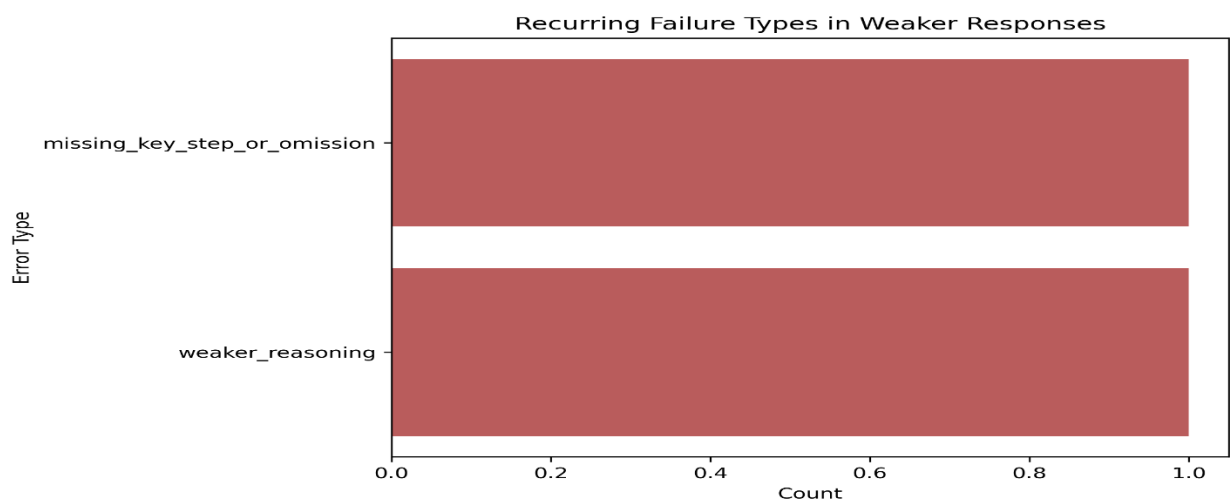
Chart 2: Average Quality Dimension Gap Between Responses

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Insight: **Clarity** and **completeness** show the largest average gaps between responses, making them the primary drivers of preference decisions. In contrast, **accuracy, relevance, consistency, and reasoning quality** show minimal differences, indicating that responses are generally aligned in correctness but differ in how clearly and fully they are presented.

Chart 3: Response Error Taxonomy



Insight: The weaker responses are most frequently characterized by **missing key steps** and **weaker reasoning**, rather than factual errors. This suggests that response quality issues are primarily due to **incomplete explanations and shallow logic**, not incorrect or irrelevant answers.

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Chart 4: Top Closest Cases (Tie/Disagreement)

query	response_1	response_2	r1_total	r2_total	preferred_response	score_gap
1. Go to a travel agent or website that specializes in flight. Choose your destination and travel dates. 2. Tell the travel agent what you want for available flights on airline websites or	1. Go to a travel agent or website that specializes in flight. Choose your destination and travel dates. 2. Tell the travel agent what you want for available flights on airline websites or	1. Go to a travel agent or website that specializes in flight. Choose your destination and travel dates. 2. Tell the travel agent what you want for available flights on airline websites or	4.166666666666667	3.5	response_1	0.666666666666667
Identify the steps to identify input elements.	Identify the steps to identify input elements.	Got it. Thanks for the clarification.	nan	nan	tie	nan
What is the difference between the colors in the ITU-R Recommendation ITU-R BT.709-6?	What is the difference between the colors in the ITU-R Recommendation ITU-R BT.709-6?	What is the difference between the colors in the ITU-R Recommendation ITU-R BT.709-6?	nan	nan	tie	nan
Here is a suggested task for the student.	Here is a suggested task for the student.	Here is a suggested task for the student.	nan	nan	tie	nan
program that query Noble price json data	program that query Noble price json data	program that query Noble price json data	nan	nan	tie	nan
get the headers of the program that reads a CSV file and displays their data	get the headers of the program that reads a CSV file and displays their data	get the headers of the program that reads a CSV file and displays their data	nan	nan	tie	nan
computer. What is the best way to add texture and flavor to your computer. Here are the steps you will	computer. What is the best way to add texture and flavor to your computer. Here are the steps you will	computer. What is the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan
might be the best way to add texture and flavor to your computer. Here are the steps you will	might be the best way to add texture and flavor to your computer. Here are the steps you will	might be the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan
to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan
to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan
to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan
to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	to be the best way to add texture and flavor to your computer. Here are the steps you will	nan	nan	tie	nan

Insight: The closest-case examples show **very small score gaps and frequent tie outcomes**, confirming that many response pairs are difficult to distinguish in terms of overall quality. The presence of **noisy or low-quality text in some responses** further contributes to weak differentiation between alternatives.

Overall visual takeaway:

Across all visualizations, the results indicate that while responses are generally correct and relevant, **differences in clarity and completeness—not correctness—are what determine preference**, and overall response differentiation remains limited.